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PROGRAMMING LANGUAGE II - PYTHON

MIT SCHOOL OF COMPUTING

AS PART OF CURRICULAM

FIRST YEAR (SEM-2)

COMPUTER SCIENCE AND ENGINEERING

UNDER THE GUIDANCE OF
Sachin Tiwari

FOR THE ACADEMIC YEAR 2025-26

SUBMITTED BY

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SOC- 9 , Enrollment no. ADT25SOCB1517

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Virtual Labs

Arithmetic Operations

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Analysis of Arithmetic Operators

Problem?

Addition

10

5

Start

Next

Reset

Step Execute

```
a = int(input('Enter a Number'))
b = int(input('Enter a Number'))
c = a + b
print("The Addition is :",c)
```

Result

a :	10
b :	5
c :	15

Output :

The Addition is : 15

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Arithmetic Operations

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Analysis of Arithmetic Operators

Problem?

Subtraction

15

3

Start

Next

Reset

Step Execute

```
a = int(input('Enter a Number'))
b = int(input('Enter a Number'))
c = a - b
print("The Subtraction is :",c)
```

Result

a :	15
b :	3
c :	12

Output :

The Subtraction is : 12

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Analysis of Arithmetic Operators

Problem?

Multiplication

10

9

Start

Next

Reset

Step Execute

```
a = int(input('Enter a Number'))
b = int(input('Enter a Number'))
c = a * b
print("The Multiplication is :",c)
```

Result

a :	10
b :	9
c :	90

Output :

The Multiplication is : 90

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Arithmetic Operations

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Analysis of Arithmetic Operators

Problem?

Division

12

2

Start

Next

Reset

Step Execute

```
a = int(input('Enter a Number'))
b = int(input('Enter a Number'))
c = a / b
print("The Division is :",c)
```

Result

a :	12
b :	2
c :	6

Output :

The Division is : 6

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What is the output of this expression,2**1**2*2**3?

- (selected) a: 16
- b: 32
- c: 34
- d: Error

The equal sign (=) is:

- a: Boolean operator
- b: Comparison operator
- (selected) c: Assignment operator
- d: Logical operator

Which one of the following have the same precedence?

- a: * and /
- b: + and -
- (selected) c: Both a and b
- d: None of the mentioned

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Built-in Functions

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Analysis of Function by ID

Function by Type

Demonstration

- **Demonstrate the working of 'id' functions**

Choose first

Integer

5

Start

Next

Reset

Step Execute

```
def identity(a):
  c = id(a)
  print(c)
```

Result(identity)

c : 59842596

Output :

59842596

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Built-in Functions

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Analysis of Function by ID

Function by Type

Demonstration

- **Demonstrate the working of 'id' functions**

Choose first

Float

5.5

Start

Next

Reset

Step Execute

```
def identity(a):
  c = id(a)
  print(c)
```

Result(identity)

c : 28255178

Output :

28255178

25°C Clear

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Built-in Functions

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Analysis of Function by ID

Function by Type

Demonstration

- **Demonstrate the working of 'id' functions**

Choose first

String

hello

Start

Next

Reset

Step Execute

```
def identity(a):
  c = id(a)
  print(c)
```

Result(identity)

c : 59036423

Output :

59036423

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- a: 1
- b: -1
- c: 2
- d: 0

What will be displayed by 'helloworld'.replace('l', 'e')?

- a: heelowored
- b: heeeoworld
- c: hllloworld
- d: heeeowored

What will be displayed by min(max(False,-2,-6), 3,5,6) ?

- a: 2
- b: -3
- c: False
- d: -4

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Loops

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Analysis of Factorial

Fibonacci Series

Prime numbers

Problem?

- **To find the Factorial of a Number.**

5

Submit

Start

Reset

Step Execute

```
for num in range(start, end + 1):
  factorial = 1
    if num < 0:
      print("Factorial for negative numbers does not exist!!!")
    elif num == 0:
      print("The factorial of the Number is :")
    else:
      for i in range(1,num + 1):
        factorial = factorial*i
      print(The Factorial of the Number is :,num)
```

Result

The Factorial of the Number is :

120

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References

Feedback

- a: key
- b: key and value in tuple format
- c: key and value in list format
- d: value

Which of the following function returns a randomly selected element from range?

- a: choice(seq)
- b: randrange([start,stop,step])
- c: random()
- d: seed()

Which of the following keyword is a valid placeholder for body of the function ?

- a: continue
- b: break
- c: pass
- d: body

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Data Types

Rate Me

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Analysis of Data Types

Problem?

Demonstrate the working of data types

Choose first

Lists

5

7

9

Start

Next

Reset

Step Execute

```
list = []
list[0] =    5
list[1] =    7
list[2] =    9
print(list)
```

Result

```
list :
[]
            [5]
            [5,7]
            [5,7,9]
Output :    [5,7,9]
```

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Data Types

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Analysis of Data Types

Problem?

Demonstrate the working of data types

Choose first

Tuples

6

3

8

Start

Next

Reset

Step Execute

```
tuple = ()
tuple[0] =    6
tuple[1] =    3
tuple[2] =    8
print(tuple)
```

Result

```
tuple :
()
                    (6)
                    (6,3)
                    (6,3,8)
Output :            (6,3,8)
```

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Data Types

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Analysis of Data Types

Problem?

Demonstrate the working of data types

Choose first

Dictionary

4

1

7

Start

Next

Reset

Step Execute

```
dict = {}
dict[0] =    4
dict[1] =    1
dict[2] =    7
print(dict)
```

Result

dict :

{}

{0 : 4}

{0 : 4, 1 : 1}

{0 : 4, 1 : 1, 2 : 7}

Output : {0 : 4, 1 : 1, 2 : 7}

25°C Clear

Search

ENG IN

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References
Feedback

Suppose list1 is [4, 5, 8], what is sum(list1) ?

- (selected) a: 17
- b: 13
- c: 15
- d: Error

Which of the following data types are immutable?

- (selected) a: Tuples
- b: Sets
- c: Dictionary
- d: Lists

Which of the following commands will create a list?

- a: mylist = list()
- b: mylist = []
- c: mylist = list([1, 2, 3])
- (selected) d: all of the mentioned

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String

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Analysis of Strings

Concatenate | Slicing | Reverse | Repetition

Demonstration

Demonstrate the position of a character by defining proper index.

python

2

Start

Next

Reset

Step Execute

```
s = input("Enter the String :")
start = input("Starting Index :")
print("The Indexed String is :",s[start])
```

Result

The Indexed String is :

t

Very humid
Now

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- a: +
- b: *
- c: -
- d: All of the mentioned

What is the output of 'virtual'+1+2+'python'?

- a: vitual3python
- b: Error
- c: virtual12python
- d: virtualpython

5 If y='123' what will be the the return type of type(y)?

- a: str
- b: int
- c: bool
- d: String

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